

BrickAtlas LDraw-1: Source-Faithful Brick Models with Numbered Visual and Structural Questions

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Abstract

An assembly benchmark needs identifiable parts and trustworthy geometry before its reasoning scores can be interpreted. We present BrickAtlas LDraw-1, a replacement dataset built from 24 unmodified LDraw/OMR source models, containing 15,334 part instances and 617 questions. Each source instance receives a stable number shared by the interactive renderer, public visual inputs, task operands, scorer, and human-review records. Questions use source attributes, author STEP metadata, recognized connector graphs, or explicitly scoped scene edits. A connector and inset-mesh audit reports its coverage and unresolved intersection candidates without asserting physical stability. Source reparsing reproduces every retained instance identity, color and pose; all reference answers pass their contracts. These are software controls, not measurements of model reasoning. We provide the original source locks, exclusions, pairwise similarity screens, numbered inputs, four-view appendix, and a review workflow. The release retires the previous synthetic geometry and its physical conclusions; model performance and human difficulty remain unmeasured.

1. Introduction

Visual plausibility is an insufficient foundation for evaluating assembly reasoning. A row of blocks may resemble a mechanism while having no real mating interfaces. Several near-identical layouts may inflate a dataset’s size without adding structural variety. A prompt that names a “support” may be impossible to ground when several pieces share that description. Each issue can invalidate an apparently precise score before the model makes a reasoning error.

These failures motivated a complete replacement of the earlier BrickAtlas Hierarchy-3 release. Human inspection identified disconnected imitation geometry, repeated construction patterns, and ambiguous module references. Its nominal servo checks did not establish real LEGO connectivity. We therefore retire that release as the active benchmark and do not transfer its counts, physical passes, or pilot

model scores to LDraw-1.

The replacement fixes the source object and authors questions around the evidence actually available. We retain complete original OMR files, their authors and licenses, and their dependency closures. Every physical part instance in the source has a stable B0001-style number. The number identifies an instance, not a reusable part type: two identical bricks receive different numbers. Relevant numbers appear with leader lines in the 3D view and in exported visual inputs. An instance can also be isolated, rotated, and inspected at its source pose.

The task contract separates four kinds of evidence. Visual questions require a numbered rendering. Source-data questions supply the relevant attributes or instruction records. Graph questions reason about recognized connector edges. Scene-edit tasks reveal an original instance or replay an author’s steps. None of these contracts establishes forces, friction, stability, or an insertion trajectory.

Our contributions are a source-faithful replacement corpus, a common numbering interface for observation and evaluation, and reproducible checks that expose the limits of structural evidence. We release a reviewable dataset and protocol rather than a model leaderboard.

2. Related Work

Interactive and generative assembly. Break and Make [10] studies interactive structural understanding through disassembly and reconstruction. Visual manual translation [11] connects assembly instructions with executable plans. BrickNet [4] provides a graph-backed representation of brick interfaces, while BrickGPT [7] studies physically stable and buildable generation. LDraw-1 uses the public BrickNet connector implementation as an audit tool for fixed original sources. It does not reproduce the generative or robotic evaluation protocols of those works.

Visual and physical reasoning. CLEVR [3] and GQA [2] motivate explicitly compositional questions with inspectable answers. BLINK [1] highlights limitations



Figure 1. **Unmodified source assemblies.** Actual 3D renders of retained OMR models, including articulated vehicles, creatures, architecture, a motorcycle, and a launch vehicle. Geometry comes from original LDraw parts and source transforms. The complete supplement provides four views and attribution for every source. No geometry is generated to increase the model count.

Setting	Primary evaluation object
Break and Make	Interactive structural reconstruction
BrickNet	Connector graphs and generated assemblies
BrickGPT	Text-driven, buildable brick generation
LEGO-Puzzles	Multi-step spatial reasoning
PhyBlock	Physical understanding and planning
LDraw-1	Fixed original sources, numbered questions, explicit source and graph evidence

Table 1. Related settings differ in their observations, actions, and success criteria. This is a task comparison, not a performance ranking.

in basic visual perception. LEGO-Puzzles [8] and PhyBlock [6] evaluate spatial and physical planning under their respective observation and simulation contracts. Our contribution concerns source authenticity, instance grounding, and the boundary between CAD evidence and physical claims.

3. Source Corpus

3.1. Acquisition and Identity

LDraw represents a reference by a part name, color, translation and linear transform [5]. An MPD file may contain nested submodels and custom parts. We resolve the complete dependency closure and flatten the source hierarchy into part instances while retaining each instance’s path, source file and line. Three.js LDrawLoader produces the

real part meshes [9]. A single documented conversion maps LDraw units and axes to the viewer’s millimeter coordinate frame.

We preserve the original MPD bytes and their SHA256 digest. The website provides the unchanged source file, author, license, and source URL. The stored scene transform, color and part type are compared with a fresh source parse for every instance. Mesh chunks and dependencies have separate checksums. Grouping for display has no role in the connector ground truth.

3.2. Selection and Exclusions

The preparation inventory contains 43 candidates from the existing source catalog and newly acquired OMR files. Of these, 29 complete sources were converted. Fourteen were excluded during preparation for explicit reasons such as unresolved or unsupported geometry, attribution gaps, oversized models, singular transforms, or a mismatch between source instances and loader identities. Five additional sources were excluded after structural auditing: duplicate placements or connector-definition coverage below 90%. The retained collection contains 24 sources from 24 set families. The machine-readable release manifest preserves every exclusion reason.

The 90% threshold limits the amount of unsupported input to graph questions. It is not a connectivity or quality score. Unknown parts remain in the displayed source and are reported explicitly; they are never deleted to raise coverage. A source scene can legitimately contain separate objects, such as a vehicle and a figure. We do not invent edges between them.

Band	Sources	Instances	Tasks
D1	5	501	83
D2	9	2172	175
D3	2	1440	59
D4	8	11221	300

Table 2. Measured source scale. D1: at most 150 instances; D2: 151–400; D3: 401–1,000; D4: more than 1,000. These bands do not assert calibrated human or model difficulty.

3.3. Diversity Screens

Only one primary build is retained per set family. Alternative builds, recolors and generated variants do not increase the source count. We compute a geometry fingerprint after removing source labels, colors, instance ordering and global translation. The retained sources have distinct fingerprints.

We additionally report all 276 source-pair comparisons. Inventory similarity is a multiset Jaccard coefficient,

$$J(A, B) = \frac{\sum_p \min(c_A(p), c_B(p))}{\sum_p \max(c_A(p), c_B(p))}, \quad (1)$$

where $c_A(p)$ is the number of instances of type p in source A . The maximum observed value is 0.450, between the large tractor and jet. Shared Technic inventories can produce this overlap even for distinct objects. A centered, scale-normalized nearest-center shape score is also supplied as a candidate screen. It is sensitive to orientation, sampling density and shared outlines; neither screen proves statistical independence. Human review remains necessary.

4. Structural Evidence

4.1. Connector Matching

We apply BrickNet 0.1.0 connector matching to the unchanged source placements, including its alias definitions and upstream tolerances. The algorithm recognizes stud/hole, fixed, axle, hinge and ball interface families. A graph vertex is a numbered source instance; an edge records the two connector indices, recognized family, and associated orientation or slide parameters. Multiple connector records can link the same pair. Graph questions use an undirected simple adjacency relation.

Coverage measures whether a part type has a connector definition. Connected components describe the graph that the matcher recognized. Neither quantity is the probability that the real assembly is connected. Custom flexible pieces, decorative parts and some printed variants can lack definitions. Alias matching and tolerances can also miss or overstate a contact.

The upstream parser normalizes rotation matrices during connector inference. We record its maximum matrix deviation per source. The original source matrices remain the

rendering ground truth. Inset-mesh collision queries use the original alias-transformed matrices, not the normalized graph poses.

4.2. Intersection Candidates

Inset collision meshes provide pairwise intersection candidates. Intersections at friction pins, gears and other mating geometry are not automatically defects. We report both all detected intersections and those without a recognized mating interface. The latter total 573 pairs across retained sources and remain visible in the review evidence. We neither suppress them nor move source parts to remove them.

This audit does not measure forces, masses, friction, material deformation or passive dynamics. An accepted source is an authentic, reproducibly imported CAD object, not a certified physical assembly. Resolving the remaining candidates requires connector-aware inspection, more complete geometry definitions, or physical validation.

5. Numbered Tasks and Observations

5.1. Instance Grounding

Let $M = \{(b_i, p_i, c_i, T_i)\}_{i=1}^n$ denote the source instances: b_i is a stable instance ID, p_i a part type, c_i a color, and T_i the original pose. The displayed number $\ell_i = \mathbb{B}||\text{pad}_4(i)$ maps to exactly one b_i within a source hash. A globally unambiguous reference is therefore $(\text{sourceHash}, b_i)$.

A question binds every operand to both b_i and ℓ_i . Leader lines anchor numbers at projected geometry centers. A label is an identity pointer, not proof that the part is unoccluded. The interface provides individual isolation and numbered inventory search. Dense tasks show at most twelve labels simultaneously, and every operand can be inspected separately. Screenshot export composites the same overlay with the model canvas.

5.2. Task Applicability

The release contains eleven families (Table 3). Tasks are generated only when their required evidence exists. Visual type matching requires another instance of the same source part type and different-type distractors. Source-step tasks require author STEP provenance. An editorial grouping or file-list order never substitutes for author instructions.

Visual tasks include color identification and part-type correspondence. Their public records omit part names, color attributes and solutions. Each of the 140 visual questions has a numbered operand image, and interactive evaluation can rotate, zoom or isolate the source. Coordinate questions explicitly supply geometry bounding-box centers; these are not centers of mass.

Connector questions classify supplied records, identify all direct neighbors, and count components after deleting a numbered vertex from an induced local graph. Deletion is

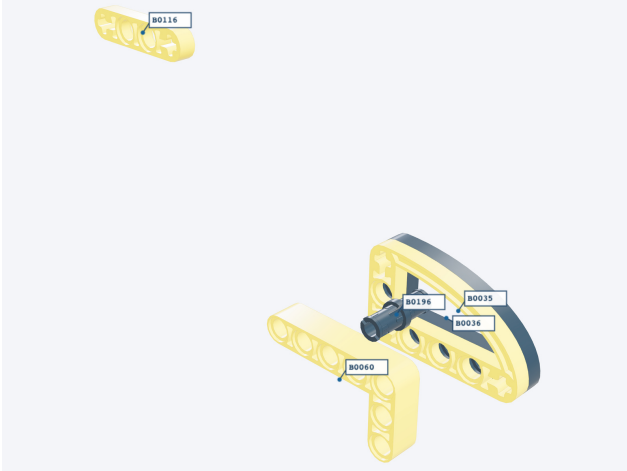


Figure 2. **Explicit instance references.** A visual correspondence question isolates its numbered operands from an original Technic source. Labels identify instances even when part types repeat. Isolation changes visibility only.

Family	N	Input
Color identification	73	Visual
Unsupported-instance evidence	17	Source data
Nearest instance	73	Source data
Limits of stability evidence	24	Source data
Components after deletion	73	Graph
Connector record classification	87	Graph
Direct connector neighbors	73	Graph
Restore numbered instance	58	Scene edit
Part-type correspondence	67	Visual
Author step sequence	17	Scene edit
Author step lookup	55	Source data

Table 3. Actual task applicability in LDraw-1. Counts vary by source; there is no universal task template quota. Record-classification and evidence-limit questions are basic controls, not claims of advanced physical reasoning.

a graph operation. It does not mean that the corresponding physical brick can be extracted without moving its neighbors. Coverage and evidence-limit questions ask what can actually be concluded from a partial audit.

5.3. Source Steps and Scene Edits

Seventeen retained sources provide usable author STEP records. The viewer replays their cumulative visible instances at the original poses. It does not interpolate fictional insertion paths. Sources without usable author steps have no replay or source-order questions.

The restoration family temporarily hides a numbered instance in a display copy. A correct action restores that instance at its original pose and preserves the remaining visibility. This is an editor task with a precise identity target.

Control	N	Rate (%)
Reference solutions	617	100.0
Public-precondition solver	75	100.0
Invalid controls rejected	1606	100.0
Always A (single choice)	469	26.7
Uniform choice (expected)	469	27.3

Table 4. Software and choice controls. Reference and invalid-control rows test contracts. The precondition solver uses public actions. The final two rows use only single-choice tasks. None is an MLLM result.

The source-sequence family reveals a short author-step window in order. It evaluates adherence to given source instructions, not a newly inferred mechanical assembly order.

6. Scoring and Review

Choice tasks use exact option IDs; set-valued outputs require the exact set with no duplicate IDs. Operation tasks use a public transition system. Each action has required facts, forbidden facts, additions, deletions and cost. The interpreter rejects unknown actions, violated preconditions, repeated completed steps and budget overruns. Success requires all goal facts and no prohibited facts.

The same interpreter drives the website’s operation replay. A rejected operation cannot silently update the visual state. Reference solutions are available in explicitly labeled review artifacts; model-facing JSON excludes them. Because this is a public development corpus, the review assets are not a secure hidden test set. Evaluation harnesses must provide only the designated inputs and log tool access.

Reviewers mark each task pass or fail. Failure requires a reason. Records include source hash, task ID, numbered operands, decision and timestamp. Browser storage uses a new LDraw-1 namespace. Export and import support batch corrections without overwriting historical human feedback. Approval is a reviewer judgment; no automatic validation step writes a human pass.

7. Verification and Evaluation Status

Independent source reparsing reproduces 15,334 instance records across all 24 sources. Checks cover identity, hierarchy, part type, color, transforms, dependency closure and mesh checksums. All 617 reference answers are accepted. A public-precondition solver satisfies all 75 scene-edit tasks. A total of 1,606 invalid choices or action sequences are rejected. Number mappings and visual input schemas are checked separately.

All 24 sources load their full instance count in the browser. Canonical perspective, front, side and top renders and numbered example views are captured from the same

application. Browser tests exercise instance isolation, explosion, restoration, author-step playback, review persistence, JSON round trips and mobile controls.

These controls are necessary for reproducibility but do not establish dataset difficulty or real-world validity. We have not run a new language-model or vision-language-model evaluation on LDraw-1, and make no comparative model-performance claim. The earlier synthetic pilot is excluded entirely.

Proposed evaluation. A future frozen evaluation should separate image-only, interactive visual, source-data and graph inputs. It should report exact success, format validity, tool usage, and source-macro averages. Splits must respect set families and inspect high-similarity source pairs. Human audits should report agreement and error categories. Instance-number permutation and option-order controls can detect shortcuts. These experiments remain proposed, with no filled-in model accuracy cells.

8. Limitations and Release Status

The corpus is small and has uneven source scales. Several task families are deliberately elementary evidence controls. Parts within one source are not independent samples, and common LEGO inventories may span sources. Familiar commercial sets may occur in model pretraining. Public reference artifacts make contamination possible.

Original OMR geometry is stronger evidence than imitation blocks, but it is still CAD authored by contributors. Unsupported connectors, intersection candidates and source conventions require review. Our 90% coverage threshold is an operational selection rule, not an empirically calibrated physical threshold. The source-step order may differ from official instruction-book pages or a feasible robot plan.

LDraw-1 is therefore a source-faithful review release. Its current contribution is a verifiable object-and-question interface with explicit limits. Claims about stability, forces, coupled motion, insertion or extraction will require additional evidence before becoming benchmark ground truth.

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